ONE-PAGE BRIEF

Post-quantum cryptography review: aptos-core @ 81d89f428fe8

Client: Aptos Labs — public repository (passive review) · Source: github.com/aptos-labs/aptos-core, commit `81d89f428fe8` · Reviewed: 2026-11-06 16:20 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 2 of 4

post-quantum in use outside the critical paths

FINDINGS

0 Critical **0** High **9** Medium **0** Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

WHAT MATTERS MOST

Medium	QSI-2026-19659	**Quantum-vulnerable BLS12-381: transaction and message signing (2 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-19660	**Quantum-vulnerable Ed25519: transaction and message signing (10 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-19662	**Quantum-vulnerable X25519: channel key exchange (TLS, Noise, handshakes) (2 file(s))** preliminary What to do: Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols).
Medium	QSI-2026-19663	**Quantum-vulnerable ECDSA: transaction and message signing (3 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...

5 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium channel key exchange (TLS, Noise, handshakes)	X25519	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)
Medium transaction and message signing	Ed25519, ECDSA, BLS12-381, BN254 (alt_bn128)	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	BN254 (alt_bn128), Ed25519, X25519, RSA	Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS...

Static review of source code and manifests; files analysed: 336. No requests were sent to the owner's systems.

Full report, content SHA-256: `496bbf2c0ceaa6407d995b8152ee8fd44f8f95ed1f641fbd08068a5bb313751f`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.